

Jeremy Zhengqi Huang

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Research Interests

My research lies at the intersection of HCI and applied ML. I design, build, and deploy human-AI systems that help people make sense of complex sensory environments and shape them around their evolving needs. My recent work spans audio information access, soundscape mediation, and personalized captions and media.

Education

- 2024- **University of Michigan** – Ann Arbor, MI
Ph.D., Computer Science & Engineering
[Soundability Lab](#), advised by [Prof. Dhruv Jain](#).
- 2021 – 2023 **University of Michigan** – Ann Arbor, MI
M.S. in Information (Specialization: Human-Computer Interaction)
Thesis: Understanding and Augmenting Debates on Reddit
Thesis advisor: Prof. Eytan Adar GPA: 3.93.
- 2016 – 2021 **The Ohio State University** – Columbus, OH
B.S. in Data Analytics and Psychology *with Research Distinction*
Thesis: Leveraging Cognitive Factors to Improve Sustainability Education
Thesis advisor: Prof. Brittany Shoots-Reinhard. GPA: 3.837.

Preprint

- P.01 **Huang, J.**, Hicks, E., Sidharth, Hayes, G.R., Jain, D. Sona: Personalized Soundscape Mediation to Support People with Sound Sensitivity. *arXiv preprint*, 2026. [arXiv:2604.00447](https://arxiv.org/abs/2604.00447). *Under Review*.

Peer-Reviewed Conference Papers

- C.06 **Huang, J.**, de Lacerda Pataca, C., Wu, L.Y., Jain D. CapTune: Caption Personalization and Tuning for Non-Speech Information in Cinematic Experiences Using Generative Models. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2025)* [Acceptance Rate: 29.7%]
- C.05 **Huang, J.**, Herskovitz, J., Wu, L.Y., Morrison, C., Jain D. Weaving Sound Information to Support Real-Time Sensemaking of Auditory Environments: Co-Designing with a DHH User. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)* [Acceptance Rate: 27.0%]
- C.04 Park, Y., Jin, A., **Huang, J.**, Carr, J., Jain D. MaskSound: Exploring Sound Masking Approaches to Support People with Autism in Managing Noise Sensitivity. In *ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2024)* [Acceptance Rate: 29.6%]
- C.03 **Huang, J.**, Wood, W, Chhabria, H, Jain D. A Human-AI Collaborative Approach for Designing Sound Awareness Systems. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)* [Acceptance Rate: 26.4%]

- C.02 Do, H., Dang, Q., **Huang J.**, and Jain D., 2023, October. AdaptiveSound: An Interactive Feedback-Loop System to Improve Sound Recognition for Deaf and Hard of Hearing Users. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2023)* (pp. 1-13)
- C.01  **Huang J.**, Chhabria, H, and Jain D., 2023, October. “Not There Yet”: Feasibility and Challenges of Mobile Sound Recognition to Support Deaf and Hard-of-Hearing People. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2023)* (pp. 1-13)

Patent

- PA.01 Dhruv Jain, Jeremy Zhengqi Huang, FNU Sidharth, *Context-Aware Sound Selection and Suppression for Noise-Sensitive People* (Pending, 64/101,380.) U.S. Provisional Patent filed by the University of Michigan.

Posters, Demos, and Extended Abstracts

- D.02 **Huang, J.**, Hicks, E., Sidharth, Hayes, G., Jain D. Demonstration of Sona: Personalizable Real-Time Soundscape Mediation for People With Sound Sensitivity. In *Adjunct Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '26)*, Detroit, MI, USA. To appear.
- D.01 **Huang, J.**, de Lacerda Pataca, C., Wu, L.Y., Jain D. Demo of CapTune: Caption Personalization and Tuning for Non-Speech Information in Cinematic Experiences Using Generative Models. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2025)*

Professional Experience

- 2021 – 2023 **Shapiro Design Lab at the University of Michigan Library** – Ann Arbor, MI
Design Researcher, Disability Culture
- Designed a technology probe investigating journey-based support and selective disclosure for people who stutter.
Mentored by Dr. Justin Schell
- 2019, 2022 **The Coca-Cola Company** – Atlanta, GA
Technical Consumer Research Intern
- Developed data pipelines and analytics interfaces supporting multi-modal (e.g., images, videos, social media posts, etc.) consumer sentiment analysis at a global scale.
- 2020-2021 **VECTOR Lab, The Ohio State University** – Columbus, OH
Research Assistant, Mentored by Prof. Jesse Fox
- Facilitated the design and evaluation of Fluid Earth, an interactive climate data visualization tool.
- 2019-2021 **Cognitive and Affective Influences in Decision Making Lab, The Ohio State University** – Columbus, OH
Research Assistant, Mentored by Prof. Brittany Shoots-Reinhard

Honors and Awards

- 2024 **University of Michigan CSE Departmental Fellowship**
- 2023 **Best Paper Honorable Mention, ASSETS 2023**
- 2023 **University of Michigan Library Research Grant**, University of Michigan

2018-2021 **Veeam Software Endowed Scholarship in Data Analytics**, Ohio State University

Service

2025– **Accessibility Chair**, The 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026), Michigan, USA

2025 **Student Volunteer**, The 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2025), Denver, CO

2024– **Reviewer**, CHI’24, CHI’25, C&C’26, UIST’26

2024– **Event Organizer**, Accessibility, HCI, and Aging Research Group, University of Michigan

2023 **Volunteer**, Michigan AI Symposium, University of Michigan

2022-2023 **Student Leadership Board**, Michigan Institute for Data Science, University of Michigan

2021 **President**, Cognitive Science Club, The Ohio State University

2018-2021 **Student Advisor**, Data Analytics Major, The Ohio State University

2020 **Mentor**, Humanities & Cognitive Sciences High School Summer Institute, The Ohio State University

Skills

AI/ML	PyTorch, Hugging Face Transformers, TensorFlow, model training and fine-tuning, multi-modal language models, quantization, edge model deployment, interactive ML
HCI Methods	Co-design, lab and field user studies, semi-structured interviews, thematic analysis, survey and experimental design, literature review
Prototyping	Python, Swift, JavaScript/TypeScript, Kotlin, SQL, C++, React, Node.js, Flask, Gradio, Figma
Languages	English, Chinese

Teaching Experiences

2025-2026 **EECS 493 User Interface Development**, Graduate Student Instructor

Academic Mentoring

2026– **Riley Ham**, University of Michigan, School of Music, Theater, and Dance (M.S.)

2024 **Yuxuan Liu**, University of Michigan, Computer Science and Engineering (Undergraduate)

2022-2024 **Hriday Chhabria**, University of Michigan, Computer Science and Engineering (Undergraduate)

2023-2024 **Wren Wood**, University of Michigan, Computer Science and Engineering (Undergraduate)

2023-2024 **Andy Jin**, University of Michigan, Computer Science and Engineering (Undergraduate)

2023-2024 **Yuni Park**, University of Michigan, Computer Science and Engineering (Undergraduate)